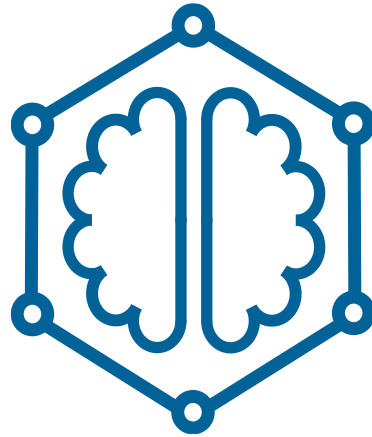




Knowledge Graphs
Organization

Teaser

Emanuel Sallinger, Markus Nissl, Aleksandar Pavlovic



Organization

1. **“Static” information:** Course Website
2. **Communication and releases:** TUWEL
3. **Mini-project** (“portfolio”)
how am I going to be marked?



Course Website

Course Content

The aim of this course is to gain a deep understanding of Knowledge Graphs divided into three blocks. An overarching aim of the course is to understand the connections between Knowledge Graphs, Artificial Intelligence, Machine Learning and Data Science.

Representations

Logical and ML-based representations for KGs.

KG Embeddings
Widely-applied, large family of ML models.

Logical Knowledge in KGs
Highly expressive, diverse family of logical models.

Graph Neural Networks
Using the KG structure as a neural network.

Data Models
Overview of data models in different communities.

[Read more details](#)

Architectures
The big picture of building IT architectures for KGs.

Scalable Reasoning
Making use of the knowledge in the KG.

KG Creation
How to create a KG from heterogeneous data?

KG Evolution
How to update, correct and complete a KG?

Systems

Systems to bring KGs into practice.

[Read more details](#)

Course Organization

COURSE ORGANIZATION

A Quick Overview

All topics will be covered by lectures offered via distance learning. Assessment is by a mini-project. The procedure is in three simple steps:

- **one-page summary** proposing a mini-project by you
- **feedback** on the proposal
- submission of the **final mini-project report** ("portfolio")

This will be supported by discussions and feedback throughout the course.

For more details on the organization of the course, including examples for mini-project ideas, exact ECTS distribution, curriculum assignment, etc., please see the below link:

[More details on organization](#)

<https://kg.dbai.tuwien.ac.at/kg-course/>



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LEARNING OUTCOMES (LOs)

Representations

The aim in this part is to understand and apply the predominant representations of knowledge and data in Knowledge Graphs.

Knowledge Graph Representations

Knowledge Graph Embeddings

Logical Knowledge

Graph Neural Networks

Knowledge Graph Data Models

Knowledge

Graph

- **(LO1)** Understand and apply **Knowledge Graph Embeddings**

Knowledge Graph Embeddings (KGEs) are a large field of representations and techniques focused on a shared principle: how to represent symbolic knowledge - as in typical databases or datasets (in our case, graphs) - in a sub-symbolic way - as in typical machine learning or deep learning scenarios (in our case, as vectors). We will cover the principles as well as selected seminal and recent KGE models such as the translation-based TransE, the semantic matching-based ComplEx, and the neural network-based ConvE.

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Course Organization Details

CURRICULA

In Which Curricula is the Course?

- E 066 931 **Logic and Computation:**
Module "Knowledge Representation and Artificial Intelligence"
- E 066 936 **Medical Informatics:**
Module "Information Processing" ("Informationsverarbeitung")
- E 066 937 **Software Engineering & Internet Computing:**
Module "Information Systems" ("Informationssysteme")
- E 066 926 **Business Informatics:**
Module "ISE/EXT – Information Systems Engineering Extension"
- E 066 645 **Data Science:**
Module "VAST/EX – Visual Analytics and Semantic Technologies - Extension"

(Update 23.02.2021: Now confirmed by both StuKos!)

COURSE DELIVERY

How Can a Portfolio Look Like?

<https://kg.dbai.tuwien.ac.at/kg-course/>



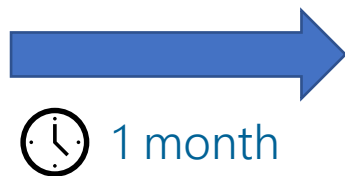
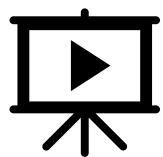
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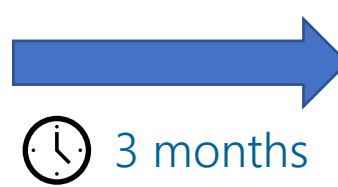


"Standard" Track

Lecture Start
March



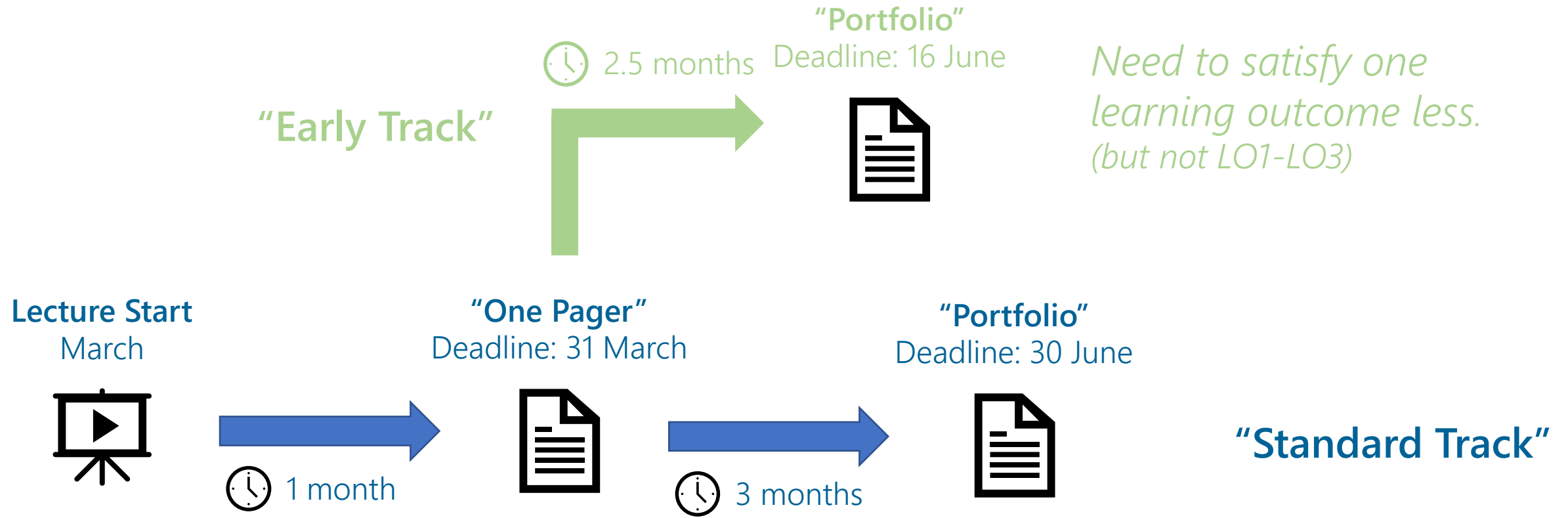
"One Pager"
Deadline: 31 March

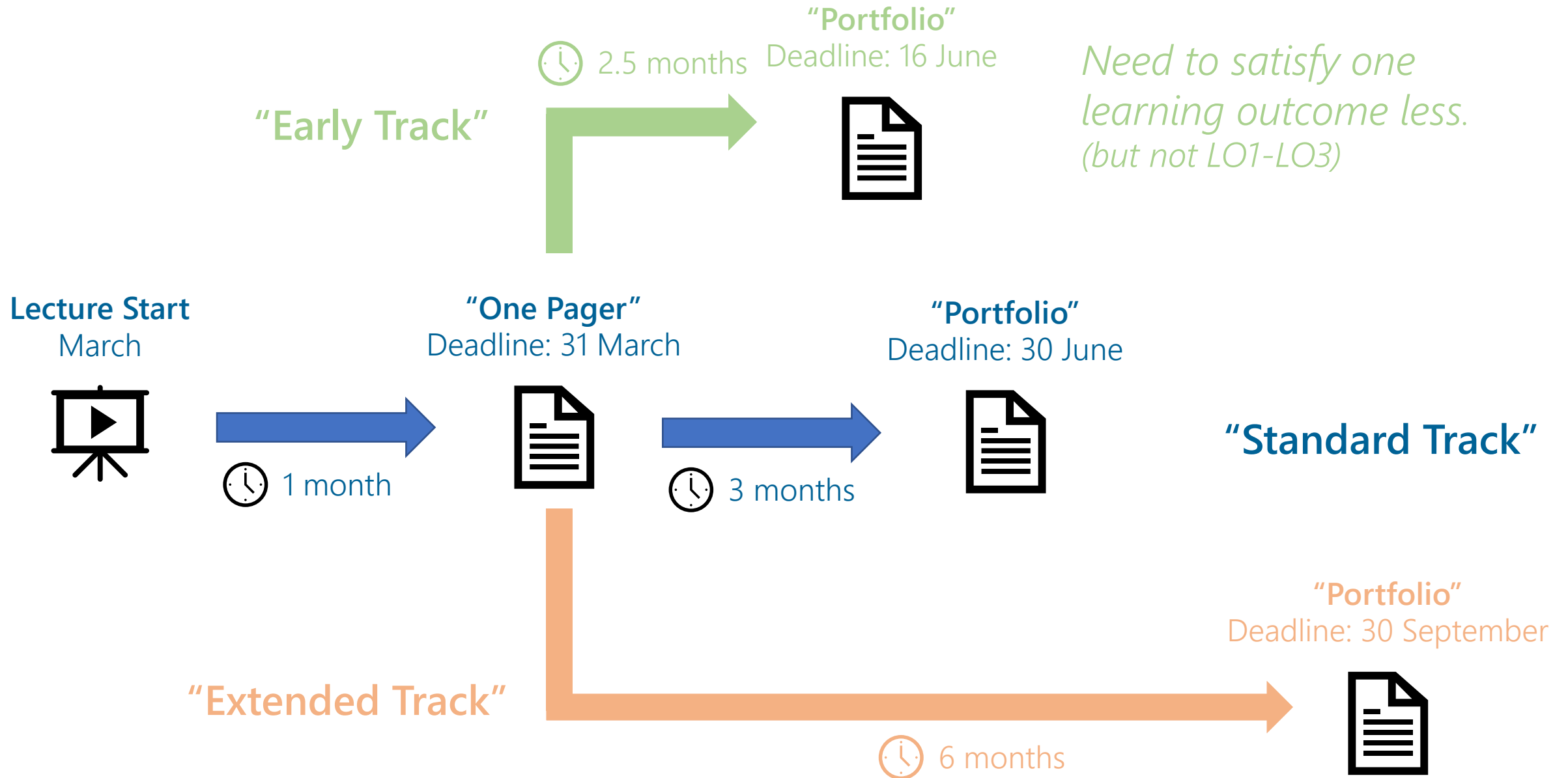


"Portfolio"
Deadline: 30 June



"Standard Track"







Organization

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how am I going to be assessed?



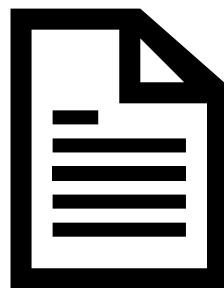
How am I going to be assessed?

By one item: the mini-project ("portfolio"):

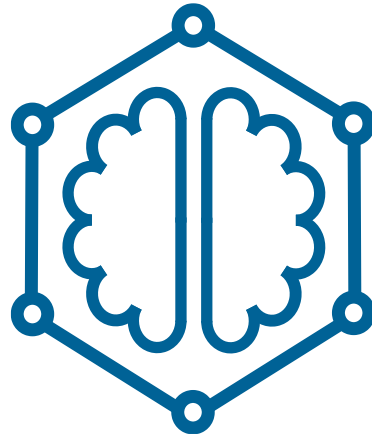
1. **one-page summary** proposing a mini-project by you
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“Portfolio”



~6 pages



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